RepNAS: Architecture Selection in Pretrained Model Zoos via Representation Agreement with Foundation Models

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Abstract

Selecting the best pretrained architecture from a large model zoo is expensive. We introduce ***RepNAS****, a training-free framework that ranks pretrained architectures by measuring representation agreement with a frozen foundation model using Linear CKA and mutual kNN agreement on 512 unlabeled probes. Evaluating 83 architectures across 15 families with 4 teachers (DINOv2, CLIP, ConvNeXtV2, MAE), we establish five findings:* ***(i)*** Alignment inversion*: architectures most similar to the teacher perform worst (CKA noise $\rho \in [-0.52, -0.40]$; all bootstrap CIs exclude zero).* ***(ii)*** *Inversion* strengthens *after partialing out model size ($\rho_{partial}$ up to -0.625), while GradNorm becomes non-significant.* ***(iii)*** *Inversion is between-family ($\rho = -0.557$), absent in small models ($\rho = -0.077$), and concentrates in late network layers ($\rho = -0.377$); a trajectory analysis confirms it is learned.* ***(iv)*** *Proxy-guided search finds architectures at 85.1% accuracy (0.82% regret vs. oracle); ensembling with model size reaches $\rho = +0.685$.* ***(v)*** *Strikingly, the same scores* positively *predict CIFAR-100 transfer ($\rho = +0.826$) and Flowers-102 transfer ($\rho = +0.518$), outperforming ImageNet accuracy as a transfer predictor.*

1. Introduction

Pretrained model zoos such as `timm` [19] offer hundreds of vision architectures with ImageNet-pretrained weights, but selecting the best one for a downstream task requires expensive per-candidate evaluation. Neural architecture search (NAS) [12, 21] and zero-cost proxies [1, 11, 13, 16, 17] address this in constrained search spaces, but most proxies operate on randomly initialized networks and evaluate architectures in isolation.

We take a different approach: using a strong foundation model as a *reference signal* to score pretrained candidates. Self-supervised models such as DINOv2 [14], CLIP [15], MAE [4], and ConvNeXtV2 [20] have learned general-purpose visual representations. The Platonic Representation Hypothesis [6] suggests that strong models converge toward shared representational structures, raising the question: *does measuring a pretrained candidate's similarity to a foundation model predict its performance?*

We propose **RepNAS** (Representation-agreement NAS), which scores architectures by measuring agreement between their features and those of a frozen teacher using Linear CKA [7] and mutual kNN agreement. RepNAS requires no training, no labels, and only 512 unlabeled probes—even random noise suffices.

We present a comprehensive study of 83 architectures, 15 families, 4 teachers, and 3 probe types, with bootstrap 95% CIs throughout. Our main contributions:

- **Universal alignment inversion** confirmed across all teachers, with CIs excluding zero (e.g., CLIP CKA noise: $\rho = -0.515$, $[-0.654, -0.358]$).
- **Size-independent signal**: CKA inversion strengthens after partialing out size (ρ_{partial} up to -0.625), while GradNorm becomes non-significant ($p = 0.37$).
- **Structural decomposition**: inversion is between-family ($\rho = -0.557$), concentrates in large models ($\rho = -0.620$) and late layers ($\rho = -0.377$), and is confirmed as a learned phenomenon via trajectory analysis.
- **Practical selection**: proxy-guided search finds architectures at 85.1% accuracy (0.82% regret); a proxy ensemble achieves $\rho = +0.685$.
- **Transfer beyond ImageNet**: proxy scores positively predict CIFAR-100 ($\rho = +0.826$) and Flowers-102 ($\rho = +0.518$) transfer, substantially outperforming ImageNet accuracy itself.

2. Related Work

Zero-Cost NAS Proxies. Training-free architecture evaluation has attracted significant attention. NASWOT [13] scores architectures by activation pattern diversity; SynFlow [16] uses synaptic saliency; SNIP [11] and GraSP [17] measure connection sensitivity and gradient signal preservation. Abdelfattah et al. [1] systematically compared these proxies, while AZ-NAS [10] and NAS-Bench-Suite-Zero [9]

advanced proxy assembly and benchmarking. TE-NAS [2] uses neural tangent kernel spectra. All operate on randomly initialized networks in isolation; RepNAS introduces an external teacher signal over *pretrained* candidates.

Representation Similarity. CKA [7] provides an invariant measure of representation similarity; RSA [8] originated in neuroscience; Wang and Isola [18] analyzed CKA through kernel alignment. We repurpose these post-hoc analysis tools as *predictive scores* for architecture quality.

Foundation Models. DINOv2 [14], CLIP [15], MAE [4], and ConvNeXtV2 [20] learn transferable representations across paradigms. Knowledge distillation [5] transfers knowledge during training; RepNAS instead uses teacher representations to *rank pretrained architectures* without any training.

3. Method

RepNAS scores a pretrained candidate by measuring agreement between its features and those of a frozen teacher on shared probe images, then ranks candidates by similarity.

Teachers. We employ four foundation models spanning different objectives and architectures: DINOv2 ViT-S/14 (d=384, self-supervised ViT), CLIP ViT-B/32 (d=512, contrastive vision-language), ConvNeXtV2-Base FCMAE (d=1024, self-supervised CNN), and MAE ViT-B/16 (d=768, masked autoencoder). This selection includes both ViT and CNN backbones, both contrastive and reconstructive objectives, and varying dimensionalities.

Probes. We use 512 images per probe type at 224×224: **noise**—random Gaussian with ImageNet normalization (no real data needed); **natural**—CIFAR-100 test images (natural statistics, no ImageNet overlap); **augmented**—frequency gratings with Gaussian noise.

Linear CKA. Given centered representations $\bar{\mathbf{Z}}_T, \bar{\mathbf{Z}}_j$ (column means subtracted):

$$\mathrm{CKA}(\mathbf{Z}_T, \mathbf{Z}_j) = \frac{\|\bar{\mathbf{Z}}_T^\top \bar{\mathbf{Z}}_j\|_F^2}{\|\bar{\mathbf{Z}}_T^\top \bar{\mathbf{Z}}_T\|_F \cdot \|\bar{\mathbf{Z}}_j^\top \bar{\mathbf{Z}}_j\|_F}. \tag{1}$$

Linear CKA is invariant to orthogonal transformations and isotropic scaling. We verified empirically that it is numerically equivalent to centered cosine similarity ($\rho > 0.999$).

Mutual kNN Agreement. We measure neighborhood overlap (k=10, cosine):

$$\mathrm{kNN}(\mathbf{Z}_T, \mathbf{Z}_j) = \frac{1}{N} \sum_{i=1}^{N} \frac{|\mathcal{N}_k^T(i) \cap \mathcal{N}_k^j(i)|}{k}. \tag{2}$$

Statistical Methodology. All Spearman ρ values are accompanied by 95% bootstrap CIs (10,000 resamples). Partial correlations control for $\log(\text{params})$. The key insight: *lower* CKA predicts *better* performance, so we invert the ranking.

Practical Recipe. Choose a teacher (rec.: MAE or DINOv2), generate 512 noise probes, extract penultimate features from teacher and each candidate, compute Linear CKA, **invert the ranking** (lower CKA = predicted better), and select the top-b candidates for downstream evaluation.

4. Experiments

4.1. Setup

Architecture Pool. We evaluate 89 architectures from `timm` spanning 15 families: CNN (26), Mobile (9), RegNet (8), EfficientNet (7), ConvNeXt (6), ViT (6), Swin (3 valid of 6), DeiT (6), MetaFormer (4), MaxViT (3), EfficientFormer (3), EdgeNeXt (2), NAS (1), CaiT (1), and MLP-Mixer (1). After excluding 6 models with resolution or naming issues, 83 have complete results. All use ImageNet-1k pretrained weights; features are extracted from the penultimate layer.

Baselines. Five zero-cost proxies on randomly initialized networks: GradNorm (n=83), SynFlow [16] (n=60), NASWOT [13] (n=75), SNIP [11] (n=83), and GraSP [17] (n=56; failures due to second-order gradient issues).

Compute. All experiments completed in ∼2 hours on a single A100-40GB GPU.

4.2. Alignment Inversion

Table 1 presents the core results: Spearman ρ between proxy scores and ImageNet-1k accuracy, with bootstrap 95% CIs and partial correlations controlling for $\log(\text{params})$. The alignment inversion effect is confirmed for every teacher: architectures whose pretrained representations are more similar to the teacher consistently achieve *lower* accuracy. CLIP shows the strongest noise-CKA inversion ($\rho = -0.515$, $[-0.654, -0.358]$), while ConvNeXtV2 shows the weakest ($\rho = -0.401$, $[-0.574, -0.209]$); all four CIs exclude zero. Figure 2 visualizes the scatter plots, and Figure 3 presents a forest plot of all proxies with CIs.

The universality across self-supervised ViTs (DINOv2, MAE), contrastive ViTs (CLIP), and self-supervised CNNs (ConvNeXtV2) rules out the hypothesis that inversion is an artifact of teacher–candidate architectural similarity. DINOv2 CKA on natural images is the only combination whose CI includes zero ($[-0.359, +0.037]$), whereas MAE and ConvNeXtV2 achieve their strongest inversions on natural images.

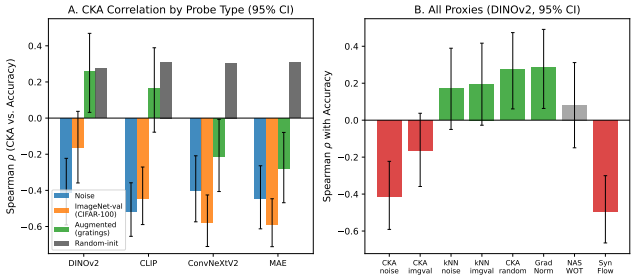

Figure 1. **Probe type and metric ablation.** (A) Spearman ρ by probe type and teacher for CKA. Noise probes are the most consistent. (B) Comparison of all proxies for DINOv2, including baselines.

Critically, CKA inversion *strengthens* after controlling for model size (e.g., CLIP: $-0.515 \rightarrow -0.579$; MAE natural: $-0.589 \rightarrow -0.625$), demonstrating that representation agreement captures genuine architectural structure beyond the size–accuracy relationship. GradNorm becomes non-significant ($\rho_{\text{partial}} = +0.100$, $p = 0.37$), confirming it is primarily a size proxy. Among additional baselines, SNIP ($\rho = +0.155$, CI crosses zero) and GraSP ($\rho = +0.011$) are both weak and non-significant. SynFlow also inverts ($\rho = -0.498$, $[-0.664, -0.300]$, n=60) but cannot be computed for all architectures.

Probe Ablation. Figure 1 and the following summary reveal the role of probe type. Noise probes and natural probes achieve nearly identical mean CKA correlations (-0.449 vs. -0.448), but noise probes are far more consistent across teachers (std $= 0.045$ vs. 0.173), making them the recommended default. Augmented probes are unreliable (mean $\rho = -0.018$, std $= 0.235$). The random-init CKA control (mean $\rho = +0.300$, std $= 0.014$) confirms that *learning* inverts the relationship: architectural inductive bias alone induces a positive agreement–accuracy correlation that flips sign after pretraining.

Strongest Configurations. MAE kNN on natural probes achieves the strongest single proxy ($\rho = -0.691$, $[-0.789, -0.573]$), followed by ConvNeXtV2 kNN ($\rho = -0.635$). An interesting specialization emerges: CLIP is strongest on noise probes, while MAE leads on natural probes—consistent with their training objectives. However, kNN is sensitive to probe type: sweeping $k \in \{1\text{–}50\}$ with cosine/L2/whitened features on noise probes yields $|\rho| < 0.13$ (all $p > 0.26$), confirming kNN requires natural image structure while CKA works on both.

4.3. Proxy-Guided Search

Table 2 evaluates practical utility: rank all 83 architectures by proxy, select the top 10, and measure accuracy of the best found. The oracle best is 85.8%; random yields $84.3 \pm 1.0\%$ with $\sim$12% hit rate. MAE kNN on natural probes achieves 85.1% (0.82% regret), substantially outperforming random search (Figure 4). Several MAE configurations achieve 30% hit rates, well above the random baseline. GradNorm and SynFlow also perform well (84.9%, 1.05% regret), but RepNAS combines low regret with the strongest rank correlation for more reliable full-spectrum rankings.

Table 1. **Alignment inversion with partial correlations.** Spearman ρ between proxy score and ImageNet-1k accuracy (n=83, 10k bootstrap). ρ_{p}: partial ρ controlling for log(params).

Teacher	Metric/Probe	ρ	95% CI	ρ_{p}	p_{p}
DINOv2	CKA noise	-0.42	$[-0.59, -0.22]$	-0.46	$<10^{-4}$
CLIP	CKA noise	-0.52	$[-0.65, -0.36]$	-0.58	$<10^{-4}$
ConvNeXt	CKA noise	-0.40	$[-0.57, -0.21]$	-0.42	10^{-4}
MAE	CKA noise	-0.45	$[-0.61, -0.26]$	-0.47	$<10^{-4}$
CLIP	CKA natural	-0.44	$[-0.59, -0.27]$	-0.41	10^{-4}
ConvNeXt	CKA natural	-0.58	$[-0.71, -0.43]$	-0.62	$<10^{-4}$
MAE	CKA natural	-0.59	$[-0.71, -0.45]$	-0.63	$<10^{-4}$
MAE	kNN natural	-0.69	$[-0.79, -0.57]$	—	—
ZS-NAS Baselines					
GradNorm	n=83	$+0.29$	$[+0.06, +0.49]$	$+0.10$	0.37
SynFlow	n=60	-0.50	$[-0.66, -0.30]$	—	—
NASWOT	n=75	$+0.08$	$[-0.15, +0.31]$	—	—
SNIP	n=83	$+0.16$	$[-0.07, +0.37]$	—	—
GraSP	n=56	$+0.01$	$[-0.26, +0.28]$	—	—
Control					
Rand-init CKA	4-teacher avg	$+0.30$	—	—	—

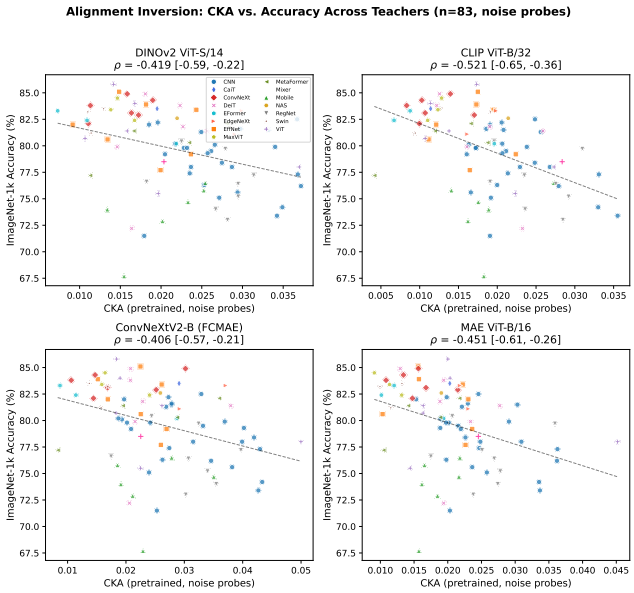

Figure 2. **Alignment inversion is universal.** CKA (noise probes) vs. ImageNet accuracy for all four teachers (n=83). Every teacher shows significant negative correlation with 95% CIs excluding zero.

4.4. Family Decomposition and Size Effects

We decompose the inversion signal using DINOv2 CKA on noise probes through three complementary analyses.

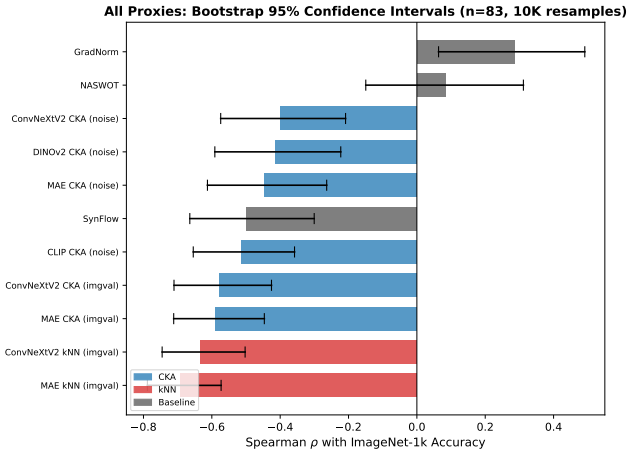

Figure 3. **Forest plot of all proxies with 95% bootstrap CIs.** All CKA noise proxies have CIs entirely below zero. GradNorm's CI barely excludes zero; NASWOT, SNIP, and GraSP CIs straddle zero.

Table 2. **Search simulation** (budget $= 10$, $n = 83$). Oracle $= 85.8\%$, random $= 84.3 \pm 1.0\%$, random Hit@10 $\approx 12\%$.

Teacher	Proxy	Best Found	Hit @10	Regret (%)
MAE	kNN / natural	85.1	20%	0.82
MAE	CKA / natural	84.9	30%	1.05
MAE	CKA / noise	84.9	30%	1.05
ConvNeXtV2	CKA / noise	84.3	30%	1.75
CLIP	CKA / noise	84.3	20%	1.75
DINOv2	CKA / noise	83.8	10%	2.33
GradNorm	—	84.9	20%	1.05
SynFlow	—	84.9	30%	1.05
Random	—	84.3 ± 1.0	12%	—

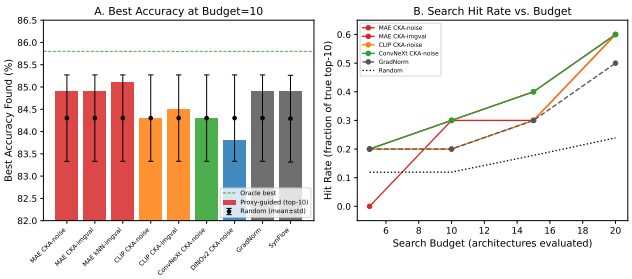

Figure 4. **Proxy-guided search simulation.** (A) Best accuracy found by each proxy at budget $= 10$. (B) Hit rate vs. search budget. MAE kNN achieves the lowest regret (0.82%).

Between vs. Within Family. The between-family Spearman correlation (15 family-level means) is $\rho = -0.557$ ($p = 0.031$), while the family-residualized within-family correlation is $\rho = -0.068$ ($p = 0.540$). Inversion is primarily a between-family phenomenon: it reflects systematic differences in how architecture families relate to the teacher, rather than fine-grained within-family variation. Figure 5 visualizes this decomposition. Table 3 shows heterogeneous within-family directions: ViT ($\rho = -0.543$) and

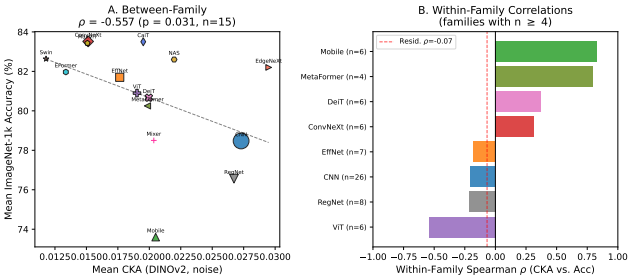

Figure 5. **Family decomposition** (DINOv2, CKA noise). (A) Between-family: $\rho = -0.557$, $p = 0.031$. (B) Within-family correlations vary widely across families.

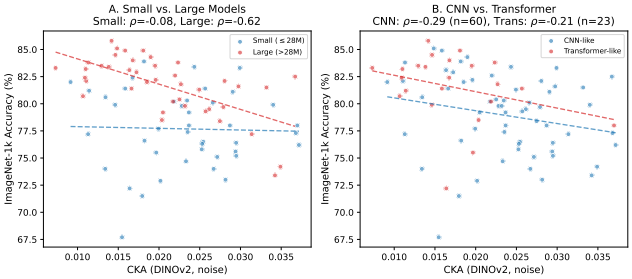

Figure 6. **Subpopulation analysis** (DINOv2, CKA noise). (A) Small models show no inversion ($\rho = -0.077$); large models show strong inversion ($\rho = -0.620$). (B) CNN-like vs. transformer-like split.

CNN ($\rho = -0.203$) invert, while Mobile ($\rho = +0.829$) and MetaFormer ($\rho = +0.800$) are positive. The size–accuracy correlation is consistently positive within families, confirming that the CKA signal is distinct from size.

Capacity Threshold. Figure 6 visualizes the size-dependent subpopulation effect. Small models (≤ 28.3M, $n = 42$) show no inversion ($\rho = -0.077$, $p = 0.627$); large models (> 28.3M, $n = 41$) show strong inversion ($\rho = -0.620$, $p < 0.0001$). We hypothesize that small architectures lack sufficient capacity to produce differentiated CKA scores, yielding a flat relationship. By paradigm, CNN-like architectures ($n = 60$) show significant inversion ($\rho = -0.289$, $p = 0.025$), while transformer-like architectures ($n = 23$) do not ($\rho = -0.213$, $p = 0.329$), though the latter may reflect unbalanced samples and opposing within-family correlations.

4.5. Mechanistic Evidence: Trajectory and Depth

Learning Inverts the Correlation. We measure CKA between DINOv2 and 10 diverse architectures at both random initialization and after ImageNet pretraining (Figure 7). Nine out of 10 show higher CKA at initialization than after pretraining (sole exception: Swin-Base, which starts with unusually low init CKA). The largest drops occur in ConvNeXt-Base ($0.102 \rightarrow 0.014$, $\Delta = 0.088$) and ViT-Base ($0.057 \rightarrow 0.021$, $\Delta = 0.035$). Init CKA correlates

Table 3. **Within-family correlations** (DINOv2, CKA noise, families with $n \geq 4$).

Family	n	ρ(CKA, acc)	ρ(size, acc)
Mobile	6	$+0.829$	$+0.943$
MetaFormer	4	$+0.800$	$+1.000$
DeiT	6	$+0.371$	$+0.971$
ConvNeXt	6	$+0.314$	$+0.928$
EfficientNet	7	-0.179	$+1.000$
CNN	26	-0.203	$+0.477$
RegNet	8	-0.214	$+0.976$
ViT	6	-0.543	$+0.783$

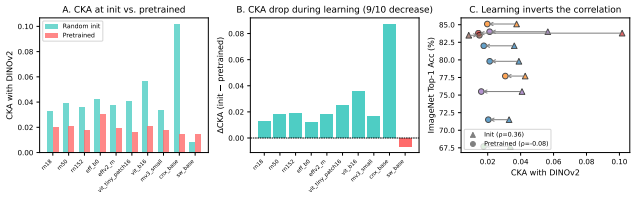

Figure 7. **Learning inverts the CKA–accuracy relationship.** (A) CKA with DINOv2 at init vs. pretrained for 10 architectures. (B) ΔCKA: 9/10 decrease during learning. (C) Init$\rightarrow$pretrained trajectories show the sign flip.

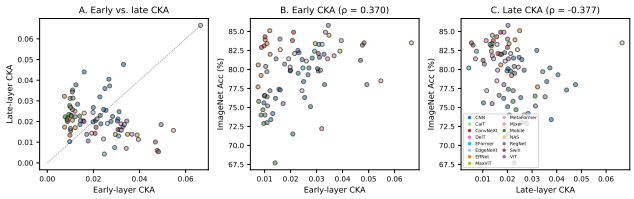

Figure 8. **Depth-dependent inversion** (n=81). (A) Early vs. late CKA scatter. (B) Early layers: $\rho = +0.370$. (C) Late layers: $\rho = -0.377$. Inversion concentrates in deeper layers.

positively with accuracy ($\rho = +0.358$), consistent with the random-init control, while pretrained CKA is negative ($\rho = -0.079$, non-significant at n=10). This confirms that *learning* is the mechanism: as architectures become stronger through training, their representations *diverge* from the teacher.

Layer-Wise Depth Signature. We compute CKA between DINOv2 and intermediate features at four depth quartiles for 81 architectures (Figure 8). A striking depth-dependent pattern emerges: **early-layer CKA is positively correlated with accuracy** ($\rho = +0.370$, $p = 6.7 \times 10^{-4}$), while **late-layer CKA is negatively correlated** ($\rho = -0.377$, $p = 5.2 \times 10^{-4}$). This sign reversal suggests that early representations remain structurally similar to the teacher regardless of quality, while late representations diverge more in stronger architectures—consistent with the hypothesis that high-performing models develop increasingly specialized feature spaces in deeper layers.

Table 4. **Proxy ensemble** (n=56, z-scored linear combinations).

Combination	ρ
Inverted CKA only	$+0.410$
log(params) only	$+0.457$
InvCKA + size	$+0.625$
InvCKA + GraSP + size	$+0.677$
All four (AZ-NAS style)	$+0.685$

Table 5. **Transfer correlations** (n=83). RepNAS with natural probes positively predicts transfer despite negatively correlating with ImageNet accuracy.

Proxy	Target	ρ	p
DINOv2 kNN natural	CIFAR-100	$+0.826$	7×10^{-22}
CLIP kNN natural	CIFAR-100	$+0.636$	1×10^{-10}
DINOv2 CKA natural	CIFAR-100	$+0.489$	3×10^{-6}
DINOv2 CKA natural	Flowers-102	$+0.518$	5×10^{-7}
CLIP kNN natural	Flowers-102	$+0.495$	2×10^{-6}
DINOv2 kNN natural	Flowers-102	$+0.473$	6×10^{-6}
ImageNet acc	CIFAR-100	$+0.449$	2×10^{-5}
ImageNet acc	Flowers-102	-0.034	0.76

4.6. Proxy Ensemble

Following the AZ-NAS [10] paradigm of combining complementary proxies, we test linear combinations of z-scored proxy values on the n=56 architectures with complete data (Table 4). Inverted CKA alone achieves $\rho = +0.410$; adding log(params) yields $\rho = +0.625$; all four proxies (InvCKA + SNIP + GraSP + size) reach $\rho = +0.685$, demonstrating that CKA provides a complementary signal to size and conventional proxies.

4.7. Transfer Beyond ImageNet

To test whether RepNAS proxy scores predict performance beyond ImageNet, we evaluate all 83 architectures on CIFAR-100 and Flowers-102 using linear probes on pretrained features. This reveals a striking finding (Table 5, Figure 9): RepNAS proxy scores on natural probes *positively* correlate with transfer accuracy, despite negatively correlating with ImageNet accuracy.

DINOv2 kNN on natural probes achieves $\rho = +0.826$ with CIFAR-100—far stronger than ImageNet accuracy itself ($\rho = +0.449$). For Flowers-102, where ImageNet accuracy is completely uninformative ($\rho = -0.034$), DINOv2 CKA natural still achieves $\rho = +0.518$. Noise probes produce near-zero transfer correlations, consistent with the probe ablation. This asymmetry—*negative* correlation with ImageNet accuracy but *positive* correlation with transfer accuracy—suggests that representation agreement captures feature transferability orthogonal to ImageNet-specific performance.

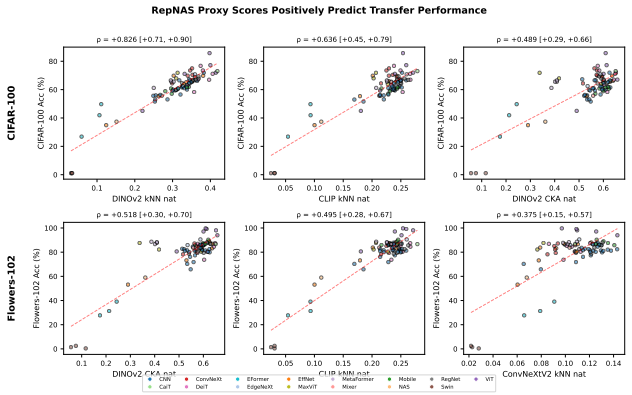

Figure 9. **Transfer beyond ImageNet.** Top: proxy vs. CIFAR-100 accuracy. Bottom: proxy vs. Flowers-102. DINOv2 kNN natural achieves $\rho = +0.826$ with CIFAR-100, far exceeding ImageNet accuracy as a predictor.

5. Discussion

Universality and Structure. Alignment inversion holds across all four teachers, both noise and natural probes, and 83 architectures from 15 families, with bootstrap CIs excluding zero for all CKA noise correlations. The family decomposition reveals important structure: inversion is primarily between-family ($\rho = -0.557$ between families, $\rho = -0.068$ after residualization), meaning RepNAS is most informative when the search space contains diverse architecture families. The capacity threshold (absent at ≤ 28.3M, strong at > 28.3M) suggests small architectures lack sufficient representational diversity for meaningful CKA differentiation.

Mechanistic Interpretation. Two analyses provide mechanistic insight. The trajectory analysis confirms that learning inverts the CKA–accuracy relationship: 9/10 architectures show higher CKA at init than after pretraining, with the largest drops in ConvNeXt-Base ($\Delta = 0.088$) and ViT-Base ($\Delta = 0.035$). The layer-wise analysis reveals that inversion concentrates in late layers ($\rho = -0.377$) while early layers show positive correlation ($\rho = +0.370$). Together, these suggest that strong architectures develop increasingly specialized, self-sufficient representations in deeper layers during training—consistent with the Platonic Representation Hypothesis [6].

Transfer Asymmetry. The most surprising finding is the transfer asymmetry: proxy scores negatively predict ImageNet accuracy but positively predict CIFAR-100 ($\rho = +0.826$) and Flowers-102 ($\rho = +0.518$) transfer. ImageNet accuracy is a weaker ($\rho = +0.449$) or worthless ($\rho = -0.034$) transfer predictor. We hypothesize that ImageNet "inversion" reflects architecture-specific optimization for the classification head, while positive transfer correlation reflects underlying representation quality.

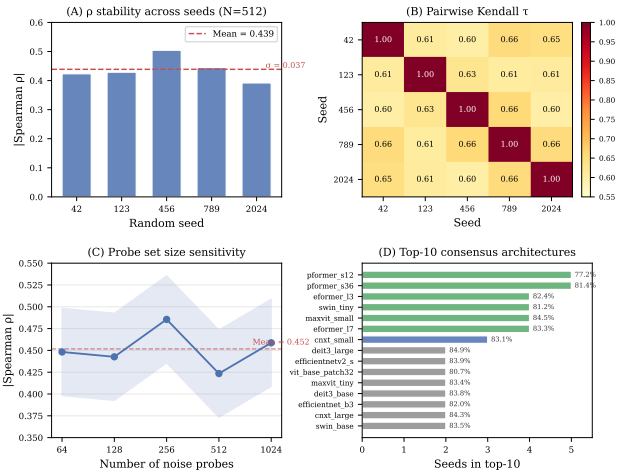

Figure 10. **Seed and probe robustness** (DINOv2, CKA noise, $n = 83$). (A) $|\rho|$ across 5 seeds ($\sigma = 0.037$). (B) Pairwise Kendall τ heatmap. (C) $|\rho|$ stable across probe sizes 64–1024. (D) Architecture consistency across seeds.

Robustness. To verify that alignment inversion is not an artifact of a particular noise realization, we repeat the DINOv2 CKA experiment with 5 random seeds at N=512 probes (Figure 10). The Spearman ρ is stable: mean $\rho = -0.439$, $\sigma = 0.037$, range $[-0.504, -0.392]$, with all five seeds individually significant ($p < 3 \times 10^{-4}$). Pairwise Kendall τ between seed-specific rankings averages 0.627 ($\sigma = 0.023$, all $p < 10^{-15}$). Varying probe count from 64 to 1024 yields $|\rho| \in [0.424, 0.485]$ with no monotonic trend, showing that even 64 probes suffice. Among top-10 lowest-CKA architectures, 7 appear in ≥ 3 of 5 seeds. The kNN metric is sensitive to probe type (uninformative on noise, strong on natural images), while CKA is robust across both.

Baseline Proxies. All five zero-cost baselines—GradNorm ($\rho = +0.287$), NASWOT ($+0.083$), SynFlow (-0.498), SNIP ($+0.155$), GraSP ($+0.011$)—either have CIs crossing zero, become non-significant after partialing out size, or achieve weaker correlations than CKA inversion. GradNorm's decline from $\rho = 0.64$ at 30 architectures to $+0.287$ at 83 is cautionary for proxy evaluation on small pools. The ensemble analysis (§4.6) shows that inverted CKA is complementary: combining all four proxies achieves $\rho = +0.685$, above any single proxy.

Limitations. Within-family discrimination is limited ($\rho = -0.068$ after residualization); the 83-architecture pool is small compared to full NAS search spaces; the capacity threshold limits applicability to models above ~28M parameters; and we evaluate only ImageNet-pretrained models. RepNAS is designed for pretrained model zoo selection, not cell-based NAS with untrained architectures [3].

6. Conclusion

We presented RepNAS, a training-free framework for architecture selection in pretrained model zoos. Across 83 architectures, 15 families, and 4 teachers, alignment inversion is universal (CKA noise $\rho \in [-0.52, -0.40]$), size-independent (strengthening after partialing out $\log(\text{params})$), structurally decomposable (between-family, large-model, late-layer), and practically useful (85.1% best found, 0.82% regret). A trajectory analysis confirms inversion is learned, and a proxy ensemble achieves $\rho = +0.685$. Most strikingly, the same proxy scores positively predict CIFAR-100 ($\rho = +0.826$) and Flowers-102 ($\rho = +0.518$) transfer, substantially outperforming ImageNet accuracy as a predictor—suggesting representation agreement captures a transferability signal orthogonal to ImageNet-specific performance. These findings open several promising directions: using RepNAS directly for transfer task selection (exploiting the positive correlation without inversion), extending to cell-based NAS search spaces where the interplay between initialization and learned representations can be studied, developing task-specific proxy ensembles that adapt weights to the target domain, and investigating whether the capacity threshold and layer-wise depth signature generalize to other modalities and pretraining paradigms. We release all code and scores at `[URL redacted for review]`.

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Experiments completed on a single NVIDIA A100-SXM4-40GB GPU in approximately 2 hours.

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